

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15B

CARBONYL COMPOUNDS (ALDEHYDES, KETONES, REDOX & NUCLEOPHILIC ADDITION)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15B: Carbonyl Compounds (Aldehydes, Ketones, Redox & Nucleophilic Addition)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	204 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Distinguishing Aldehydes and Ketones****[5 Marks]****WCH14/01/Jan23/Q17**

Propanal ($\text{CH}_3\text{CH}_2\text{CHO}$) and propanone (CH_3COCH_3) are structural isomers.

- (a) Describe the observation when propanal is warmed with Tollens reagent $[\text{Ag}(\text{NH}_3)_2]^+$. [2]
- (b) Describe the observation when propanal is warmed with Fehling solution. [2]
- (c) State why propanone gives no reaction with Tollens reagent or Fehling solution. [1]

Q2. 2. Nucleophilic Addition Mechanism of HCN**[5 Marks]****WCH14/01/Oct22/Q18**

Ethanal reacts with HCN in the presence of KCN catalyst to form 2-hydroxypropanenitrile.

- (a) Draw the mechanism for this nucleophilic addition reaction, showing all dipoles, lone pairs, and curly arrows. [4]
- (b) Explain why KCN is added as a catalyst. [1]

Q3. 3. Carbonyl Reaction & Test Step 3**[3 Marks]****WCH14/01/Sample/Q3**

Consider carbonyl compound 3 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q4. 4. Carbonyl Reaction & Test Step 4**[3 Marks]****WCH14/01/Sample/Q4**

Consider carbonyl compound 4 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q5. 5. Carbonyl Reaction & Test Step 5**[3 Marks]****WCH14/01/Sample/Q5**

Consider carbonyl compound 5 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q6. 6. Carbonyl Reaction & Test Step 6**[3 Marks]****WCH14/01/Sample/Q6**

Consider carbonyl compound 6 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q7. 7. Carbonyl Reaction & Test Step 7**[3 Marks]****WCH14/01/Sample/Q7**

Consider carbonyl compound 7 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]

- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q8. 8. Carbonyl Reaction & Test Step 8**[3 Marks]****WCH14/01/Sample/Q8**

Consider carbonyl compound 8 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q9. 9. Carbonyl Reaction & Test Step 9**[3 Marks]****WCH14/01/Sample/Q9**

Consider carbonyl compound 9 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q10. 10. Carbonyl Reaction & Test Step 10**[3 Marks]****WCH14/01/Sample/Q10**

Consider carbonyl compound 10 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q11. 11. Carbonyl Reaction & Test Step 11**[3 Marks]****WCH14/01/Sample/Q11**

Consider carbonyl compound 11 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q12. 12. Carbonyl Reaction & Test Step 12**[3 Marks]****WCH14/01/Sample/Q12**

Consider carbonyl compound 12 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q13. 13. Carbonyl Reaction & Test Step 13**[3 Marks]****WCH14/01/Sample/Q13**

Consider carbonyl compound 13 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q14. 14. Carbonyl Reaction & Test Step 14**[3 Marks]****WCH14/01/Sample/Q14**

Consider carbonyl compound 14 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/ NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
(b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q15. 15. Carbonyl Reaction & Test Step 15**[3 Marks]****WCH14/01/sample/Q15**

Consider carbonyl compound 15 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q16. 16. Carbonyl Reaction & Test Step 16**[3 Marks]****WCH14/01/sample/Q16**

Consider carbonyl compound 16 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q17. 17. Carbonyl Reaction & Test Step 17**[3 Marks]****WCH14/01/sample/Q17**

Consider carbonyl compound 17 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q18. 18. Carbonyl Reaction & Test Step 18**[3 Marks]****WCH14/01/sample/Q18**

Consider carbonyl compound 18 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q19. 19. Carbonyl Reaction & Test Step 19**[3 Marks]****WCH14/01/sample/Q19**

Consider carbonyl compound 19 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q20. 20. Carbonyl Reaction & Test Step 20**[3 Marks]****WCH14/01/sample/Q20**

Consider carbonyl compound 20 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q21. 21. Carbonyl Reaction & Test Step 21**[3 Marks]****WCH14/01/sample/Q21**

Consider carbonyl compound 21 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q22. 22. Carbonyl Reaction & Test Step 22**[3 Marks]****WCH14/01/Sample/Q22**

Consider carbonyl compound 22 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q23. 23. Carbonyl Reaction & Test Step 23**[3 Marks]****WCH14/01/Sample/Q23**

Consider carbonyl compound 23 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q24. 24. Carbonyl Reaction & Test Step 24**[3 Marks]****WCH14/01/Sample/Q24**

Consider carbonyl compound 24 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

Q25. 25. Carbonyl Reaction & Test Step 25**[3 Marks]****WCH14/01/Sample/Q25**

Consider carbonyl compound 25 reacting with 2,4-dinitrophenylhydrazine (2,4-DNPH) and iodine/NaOH (iodoform test).

- (a) State the observation when 2,4-DNPH is added. [1]
- (b) State the observation in the iodoform test if the compound contains a $\text{CH}_3\text{C}=\text{O}$ group. [2]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Carbonyl Reduction & Multi-Step Identification 26****[5 Marks]**
WCH14/01/Hard/Q26

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q27. 27. A* Challenge: Carbonyl Reduction & Multi-Step Identification 27**[5 Marks]**
WCH14/01/Hard/Q27

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q28. 28. A* Challenge: Carbonyl Reduction & Multi-Step Identification 28**[5 Marks]**
WCH14/01/Hard/Q28

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q29. 29. A* Challenge: Carbonyl Reduction & Multi-Step Identification 29**[5 Marks]**
WCH14/01/Hard/Q29

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q30. 30. A* Challenge: Carbonyl Reduction & Multi-Step Identification 30**[5 Marks]**
WCH14/01/Hard/Q30

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q31. 31. A* Challenge: Carbonyl Reduction & Multi-Step Identification 31**[5 Marks]**
WCH14/01/Hard/Q31

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q32. 32. A* Challenge: Carbonyl Reduction & Multi-Step Identification 32**[5 Marks]**
WCH14/01/Hard/Q32

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q33. 33. A* Challenge: Carbonyl Reduction & Multi-Step Identification 33**[5 Marks]**
WCH14/01/Hard/Q33

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q34. 34. A* Challenge: Carbonyl Reduction & Multi-Step Identification 34**[5 Marks]**
WCH14/01/Hard/Q34

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q35. 35. A* Challenge: Carbonyl Reduction & Multi-Step Identification 35**[5 Marks]**
WCH14/01/Hard/Q35

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q36. 36. A* Challenge: Carbonyl Reduction & Multi-Step Identification 36**[5 Marks]**
WCH14/01/Hard/Q36

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q37. 37. A* Challenge: Carbonyl Reduction & Multi-Step Identification 37**[5 Marks]**
WCH14/01/Hard/Q37

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q38. 38. A* Challenge: Carbonyl Reduction & Multi-Step Identification 38**[5 Marks]**
WCH14/01/Hard/Q38

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q39. 39. A* Challenge: Carbonyl Reduction & Multi-Step Identification 39**[5 Marks]**
WCH14/01/Hard/Q39

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q40. 40. A* Challenge: Carbonyl Reduction & Multi-Step Identification 40**[5 Marks]**
WCH14/01/Hard/Q40

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q41. 41. A* Challenge: Carbonyl Reduction & Multi-Step Identification 41**[5 Marks]**
WCH14/01/Hard/Q41

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q42. 42. A* Challenge: Carbonyl Reduction & Multi-Step Identification 42**[5 Marks]**
WCH14/01/Hard/Q42

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q43. 43. A* Challenge: Carbonyl Reduction & Multi-Step Identification 43**[5 Marks]**
WCH14/01/Hard/Q43

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q44. 44. A* Challenge: Carbonyl Reduction & Multi-Step Identification 44**[5 Marks]**
WCH14/01/Hard/Q44

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q45. 45. A* Challenge: Carbonyl Reduction & Multi-Step Identification 45**[5 Marks]**
WCH14/01/Hard/Q45

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q46. 46. A* Challenge: Carbonyl Reduction & Multi-Step Identification 46**[5 Marks]**
WCH14/01/Hard/Q46

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q47. 47. A* Challenge: Carbonyl Reduction & Multi-Step Identification 47**[5 Marks]**
WCH14/01/Hard/Q47

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q48. 48. A* Challenge: Carbonyl Reduction & Multi-Step Identification 48**[5 Marks]**
WCH14/01/Hard/Q48

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q49. 49. A* Challenge: Carbonyl Reduction & Multi-Step Identification 49**[5 Marks]**
WCH14/01/Hard/Q49

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

Q50. 50. A* Challenge: Carbonyl Reduction & Multi-Step Identification 50**[5 Marks]**
WCH14/01/Hard/Q50

An unknown carbonyl compound Z (C_4H_8O) gives an orange precipitate with 2,4-DNPH and a yellow precipitate with $I_2/NaOH$.

Z is reduced by $NaBH_4$ to form alcohol W.

- (a) Deduce the structural formula of Z and alcohol W. [3]
- (b) Write the equation for the reduction of Z using $NaBH_4$ (represented by $[H]$). [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Distinguishing Aldehydes and Ketones 1. (a) Silver mirror / grey precipitate forms (2). 1. (b) Blue solution forms a brick-red precipitate of Cu ₂ O (2). 1. (c) Propanone is a ketone and cannot be easily oxidized without breaking C-C bonds (1).	[5]
Q2	Q2: 2. Nucleophilic Addition Mechanism of HCN 2. (a) Nucleophilic attack by :CN ⁻ lone pair on delta+ C of C=O bond (1). Arrow from C=O bond onto O (1). Intermediate CH ₃ CH(O ⁻)CN formed (1). Protonation of O ⁻ by H ⁺ / HCN to form -OH (1). 2. (b) KCN dissociates fully to provide a high concentration of CN ⁻ nucleophiles (HCN is a weak acid) (1).	[5]
Q3	Q3: 3. Carbonyl Reaction & Test Step 3 3. (a) Bright orange / yellow precipitate (1). 3. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q4	Q4: 4. Carbonyl Reaction & Test Step 4 4. (a) Bright orange / yellow precipitate (1). 4. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q5	Q5: 5. Carbonyl Reaction & Test Step 5 5. (a) Bright orange / yellow precipitate (1). 5. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q6	Q6: 6. Carbonyl Reaction & Test Step 6 6. (a) Bright orange / yellow precipitate (1). 6. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q7	Q7: 7. Carbonyl Reaction & Test Step 7 7. (a) Bright orange / yellow precipitate (1). 7. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q8	Q8: 8. Carbonyl Reaction & Test Step 8 8. (a) Bright orange / yellow precipitate (1). 8. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q9	Q9: 9. Carbonyl Reaction & Test Step 9 9. (a) Bright orange / yellow precipitate (1). 9. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q10	Q10: 10. Carbonyl Reaction & Test Step 10 10. (a) Bright orange / yellow precipitate (1). 10. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q11	Q11: 11. Carbonyl Reaction & Test Step 11 11. (a) Bright orange / yellow precipitate (1). 11. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q12	Q12: 12. Carbonyl Reaction & Test Step 12 12. (a) Bright orange / yellow precipitate (1). 12. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q13	Q13: 13. Carbonyl Reaction & Test Step 13 13. (a) Bright orange / yellow precipitate (1). 13. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q14	Q14: 14. Carbonyl Reaction & Test Step 14 14. (a) Bright orange / yellow precipitate (1). 14. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q15	Q15: 15. Carbonyl Reaction & Test Step 15 15. (a) Bright orange / yellow precipitate (1). 15. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]

Q16	Q16: 16. Carbonyl Reaction & Test Step 16 16. (a) Bright orange / yellow precipitate (1). 16. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q17	Q17: 17. Carbonyl Reaction & Test Step 17 17. (a) Bright orange / yellow precipitate (1). 17. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q18	Q18: 18. Carbonyl Reaction & Test Step 18 18. (a) Bright orange / yellow precipitate (1). 18. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q19	Q19: 19. Carbonyl Reaction & Test Step 19 19. (a) Bright orange / yellow precipitate (1). 19. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q20	Q20: 20. Carbonyl Reaction & Test Step 20 20. (a) Bright orange / yellow precipitate (1). 20. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q21	Q21: 21. Carbonyl Reaction & Test Step 21 21. (a) Bright orange / yellow precipitate (1). 21. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q22	Q22: 22. Carbonyl Reaction & Test Step 22 22. (a) Bright orange / yellow precipitate (1). 22. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q23	Q23: 23. Carbonyl Reaction & Test Step 23 23. (a) Bright orange / yellow precipitate (1). 23. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q24	Q24: 24. Carbonyl Reaction & Test Step 24 24. (a) Bright orange / yellow precipitate (1). 24. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q25	Q25: 25. Carbonyl Reaction & Test Step 25 25. (a) Bright orange / yellow precipitate (1). 25. (b) Pale yellow precipitate of tri-iodomethane (CHI ₃) (1) with an antiseptic smell (1).	[3]
Q26	Q26: 26. A* Challenge: Carbonyl Reduction & Multi-Step Identification 26 26. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 26. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q27	Q27: 27. A* Challenge: Carbonyl Reduction & Multi-Step Identification 27 27. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 27. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q28	Q28: 28. A* Challenge: Carbonyl Reduction & Multi-Step Identification 28 28. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 28. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q29	Q29: 29. A* Challenge: Carbonyl Reduction & Multi-Step Identification 29 29. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 29. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q30	Q30: 30. A* Challenge: Carbonyl Reduction & Multi-Step Identification 30 30. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 30. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]

Q31	Q31: 31. A* Challenge: Carbonyl Reduction & Multi-Step Identification 31 31. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 31. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q32	Q32: 32. A* Challenge: Carbonyl Reduction & Multi-Step Identification 32 32. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 32. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q33	Q33: 33. A* Challenge: Carbonyl Reduction & Multi-Step Identification 33 33. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 33. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q34	Q34: 34. A* Challenge: Carbonyl Reduction & Multi-Step Identification 34 34. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 34. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q35	Q35: 35. A* Challenge: Carbonyl Reduction & Multi-Step Identification 35 35. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 35. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q36	Q36: 36. A* Challenge: Carbonyl Reduction & Multi-Step Identification 36 36. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 36. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q37	Q37: 37. A* Challenge: Carbonyl Reduction & Multi-Step Identification 37 37. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 37. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q38	Q38: 38. A* Challenge: Carbonyl Reduction & Multi-Step Identification 38 38. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 38. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q39	Q39: 39. A* Challenge: Carbonyl Reduction & Multi-Step Identification 39 39. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 39. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q40	Q40: 40. A* Challenge: Carbonyl Reduction & Multi-Step Identification 40 40. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 40. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q41	Q41: 41. A* Challenge: Carbonyl Reduction & Multi-Step Identification 41 41. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 41. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q42	Q42: 42. A* Challenge: Carbonyl Reduction & Multi-Step Identification 42 42. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 42. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]
Q43	Q43: 43. A* Challenge: Carbonyl Reduction & Multi-Step Identification 43 43. (a) Z gives iodoform test → contains CH ₃ C=O. Z is a C ₄ ketone → butanone, CH ₃ COCH ₂ CH ₃ (2). W = butan-2-ol, CH ₃ CH(OH)CH ₂ CH ₃ (1). 43. (b) CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ (2).	[5]

Q44	Q44: 44. A* Challenge: Carbonyl Reduction & Multi-Step Identification 44 44. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 44. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q45	Q45: 45. A* Challenge: Carbonyl Reduction & Multi-Step Identification 45 45. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 45. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q46	Q46: 46. A* Challenge: Carbonyl Reduction & Multi-Step Identification 46 46. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 46. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q47	Q47: 47. A* Challenge: Carbonyl Reduction & Multi-Step Identification 47 47. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 47. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q48	Q48: 48. A* Challenge: Carbonyl Reduction & Multi-Step Identification 48 48. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 48. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q49	Q49: 49. A* Challenge: Carbonyl Reduction & Multi-Step Identification 49 49. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 49. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]
Q50	Q50: 50. A* Challenge: Carbonyl Reduction & Multi-Step Identification 50 50. (a) Z gives iodoform test → contains $\text{CH}_3\text{C}=\text{O}$. Z is a C4 ketone → butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$ (2). W = butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (1). 50. (b) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (2).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15B

FAQ #1: Tollens Reagent Preparation and Hazard

Category: Tollens Test

COMMON EXAMINER TRAP / MISTAKE:

Heating Tollens reagent strongly or storing it.

MODEL ANSWER / EXAMINER EXPECTATION:

Tollens reagent $[\text{Ag}(\text{NH}_3)_2]^+$ must be freshly prepared by adding dilute NH_3 to AgNO_3 until Ag_2O ppt redissolves. Never heat strongly or store (forms explosive silver fulminate).

FAQ #2: Fehling Solution Color Change

Category: Fehling Test

COMMON EXAMINER TRAP / MISTAKE:

Writing 'solution turns red' instead of precipitate.

MODEL ANSWER / EXAMINER EXPECTATION:

Fehling solution starts BLUE. Upon warming with an aldehyde, a BRICK-RED PRECIPITATE of copper(I) oxide (Cu_2O) forms.

FAQ #3: Role of KCN in HCN Addition

Category: Nucleophilic Addition

COMMON EXAMINER TRAP / MISTAKE:

Omitting KCN and claiming HCN alone is sufficient.

MODEL ANSWER / EXAMINER EXPECTATION:

HCN is a weak acid with negligible dissociation. KCN is added to supply CN^- nucleophiles to initiate the attack on the carbonyl carbon.

FAQ #4: 2,4-DNPH Derivative Purification

Category: Qualitative Analysis

COMMON EXAMINER TRAP / MISTAKE:

Stating 2,4-DNPH identifies the specific aldehyde/ketone by color.

MODEL ANSWER / EXAMINER EXPECTATION:

2,4-DNPH gives an orange ppt for ALL aldehydes and ketones. To identify the specific compound: filter, purify by recrystallisation, measure its sharp MELTING POINT, and compare with data tables.

FAQ #5: Iodoform Test Positive Structures

Category: Tri-iodomethane Test

COMMON EXAMINER TRAP / MISTAKE:

Assuming ONLY methyl ketones give a positive iodoform test.

MODEL ANSWER / EXAMINER EXPECTATION:

Iodoform test ($\text{I}_2 + \text{NaOH}$) is positive for: (1) Methyl ketones $\text{CH}_3\text{C}=\text{O}$, (2) Ethanal CH_3CHO , (3) Secondary methyl alcohols $\text{CH}_3\text{CH}(\text{OH})\text{R}$, and (4) Ethanol $\text{CH}_3\text{CH}_2\text{OH}$ (oxidized in situ to ethanal).

FAQ #6: LiAlH_4 vs NaBH_4 Reducing Agents

Category: Reduction Reagents

COMMON EXAMINER TRAP / MISTAKE:

Using NaBH_4 in dry ether or LiAlH_4 in water.

MODEL ANSWER / EXAMINER EXPECTATION:

LiAlH_4 is a stronger reducing agent reacting violently with water $\rightarrow$ MUST be used in DRY ETHER.
 NaBH_4 is milder and safe in AQUEOUS ETHANOL solvent.

FAQ #7: Reduction Product Class**Category: Reduction****COMMON EXAMINER TRAP / MISTAKE:**

Assuming NaBH_4 reduces ketones to primary alcohols.

MODEL ANSWER / EXAMINER EXPECTATION:

Reduction of aldehydes with NaBH_4 yields PRIMARY alcohols (RCH_2OH). Reduction of ketones yields SECONDARY alcohols ($\text{RCH}(\text{OH})\text{R}'$).

FAQ #8: Nucleophilic Addition Curly Arrow Direction**Category: Mechanism****COMMON EXAMINER TRAP / MISTAKE:**

Drawing arrow from $\text{C}=\text{O}$ onto CN^- .

MODEL ANSWER / EXAMINER EXPECTATION:

Curly arrow MUST start from the LONE PAIR on the CN^- carbon atom and point to the δ^+ carbonyl carbon. A second arrow goes from $\text{C}=\text{O}$ double bond onto the oxygen atom.

FAQ #9: Why Aldehydes are More Reactive Than Ketones**Category: Reactivity Comparison****COMMON EXAMINER TRAP / MISTAKE:**

Attributing aldehyde reactivity solely to steric factors.

MODEL ANSWER / EXAMINER EXPECTATION:

Aldehydes are more reactive than ketones due to BOTH (1) Steric factors (only 1 bulky alkyl group blocking attack) and (2) Electronic factors (alkyl groups are electron-donating, reducing δ^+ on carbonyl carbon in ketones).

FAQ #10: Aldehyde Oxidation Products**Category: Oxidation****COMMON EXAMINER TRAP / MISTAKE:**

Stating propanal oxidizes to propan-1-ol.

MODEL ANSWER / EXAMINER EXPECTATION:

Oxidation of aldehydes produces CARBOXYLIC ACIDS (e.g. propanal $\rightarrow$ propanoic acid). Reduction produces primary alcohols.